DSA 8020 Statistical Methods II
Project II

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Due Date: April 3 (Thursday), 11:59pm ET

Problem 1 Regression Modeling: Generalized Linear Model vs Linear Regression Model

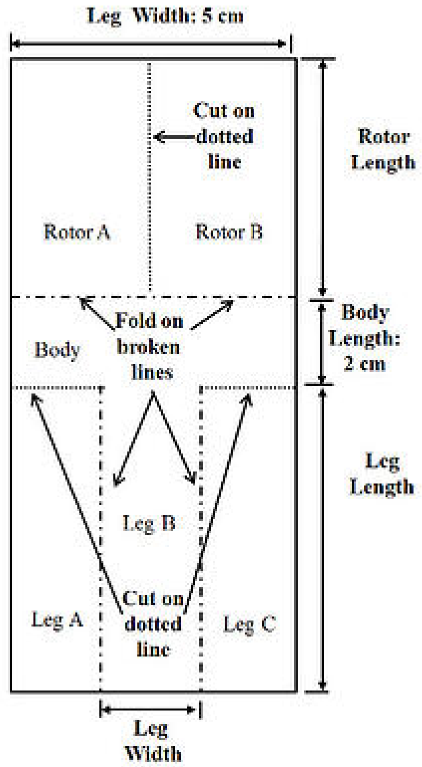
Perform a detailed regression analysis using the `gala` data set. Use `Species` as the response variable, excluding `Endemics` from this analysis, to address the following questions:

1. Conduct a regression analysis using one of the generalized linear models we covered in Week 7 (i.e., logistic regression or Poisson regression) and perform model selection.

2. Perform a linear regression analysis and conduct model selection again to determine the most appropriate linear regression model. Compare it with the GLM model selected in the previous question and comment on your findings.

Problem 2 Paper Helicopter Experiment

A researcher would like to investigate how long a paper 'helicopter' can stay in the air when dropped from a height of 2 meters. Here's how she makes the paper helicopters (see the figure below for an example) to be used for the experiment.



Source:
`https://blog.minitab.com/en/learning-design-of-experiments-with-paper-helicopters-and`

- Step 1: Cut the paper to a width of 5cm.
- Step 2: Cut the paper the length of paper rotor length plus leg length, and add 2 cm for the body.
- Step 3: Cut dotted lines at Leg A and Leg C. The length of each cut is 5 cm minus leg width divided by 2.
- Step 4: Fold legs A and B onto leg B.
- Step 5: Fold rotor A and rotor B in opposite directions. They should form 90° to the body and be 180° away from each other.

Some potential treatment factors are: 1) Paper type (light, medium, and heavy); 2) Rotor length (7.5cm or 8.5cm); 3) Leg length (7.5cm or 12cm); 4) Leg width (3.2cm or 5cm). Answer the following questions:

(a) Suppose the researcher would like to compare the responses between different paper types while keeping other factors fixed. What are the experimental units and measurement units[1] in this study? Which design can she apply?

(b) Suppose the researcher would like to conduct an experiment in (a) using 15 paper helicopters. Explain how she can perform the randomization and replication procedures here.

[1]Please don't confuse it with the other experimental unit, for example, seconds. Check the lecture slides in Week 8 for the definition of experimental unit in the context of design of experiments.

(c) Think of a situation in which the researcher would need to consider using a randomized complete block design (RCBD).

(d) If the researcher wants to study the effects of leg length and width on the flying time of heavy paper helicopters, which design is most appropriate for this situation? If the researcher wants to have 5 replicates for each treatment combination, how many heavy paper helicopters does she need?

(e) Conduct your own experiment, including designing the experiment, collecting and analyzing the data, to investigate the effects of rotor length, leg length, and leg width on the response, while keeping the paper type fixed.

Problem 3 Paper Helicopter Computer Experiment

A sophisticated computer model has been developed to study how the flying time depends on the rotor length and leg length. The output of the computer can be found in `ProjectII_prob2_data.RData` (`x1`: rotor length (cm); `x2`: leg length (cm); `y`: flying time (in seconds)). Answer the following questions:

(a) Compute and visualize the prediction surface by computing the predicted values at a dense set of grid points (`x1g <- seq(6, 10, 0.1)`; `x2g <- seq(6, 12, 0.2)`; `grids <- expand.grid(x1g, x2g)`).

(b) Compute and visualize the prediction uncertainty using a Gaussian process model. Briefly comment on your findings.